

Carbon-24 Allotropes

Stability Analysis & Energy Prediction

*Phân nhóm và hệ thống hóa các dạng thù hình Carbon
dựa trên đặc trưng hình học và mức ổn định năng lượng*

Dataset: 10,153 Carbon Structures
Clusters: 3
Best Model: Random Forest
Prediction Accuracy: $R^2 = 1.0000$

EXECUTIVE SUMMARY

PROJECT OVERVIEW

This analysis extends the initial clustering study of Carbon-24 allotropes to include:

- Stability classification of structural groups
- Machine learning-based energy prediction
- Comprehensive visualization and analysis

DATASET

- Total Structures: 10,153
- Features Used: 22
- Clustering Method: K-means (k=3)

KEY FINDINGS

1. STABILITY CLASSIFICATION

Three distinct stability groups were identified:

- Cluster 0: Highly Stable
 - Structures: 3,409 (33.6%)
 - Mean Energy: -0.4725 eV/atom
- Cluster 1: Highly Stable
 - Structures: 2,229 (22.0%)
 - Mean Energy: -0.3771 eV/atom
- Cluster 2: Less Stable
 - Structures: 4,515 (44.5%)
 - Mean Energy: 0.5429 eV/atom

2. ENERGY PREDICTION

- Best Model: Random Forest
- Test R²: 1.0000
- Test MAE: 0.0014 eV/atom
- Prediction accuracy is near-perfect, demonstrating strong structure-energy relationships

3. FEATURE IMPORTANCE

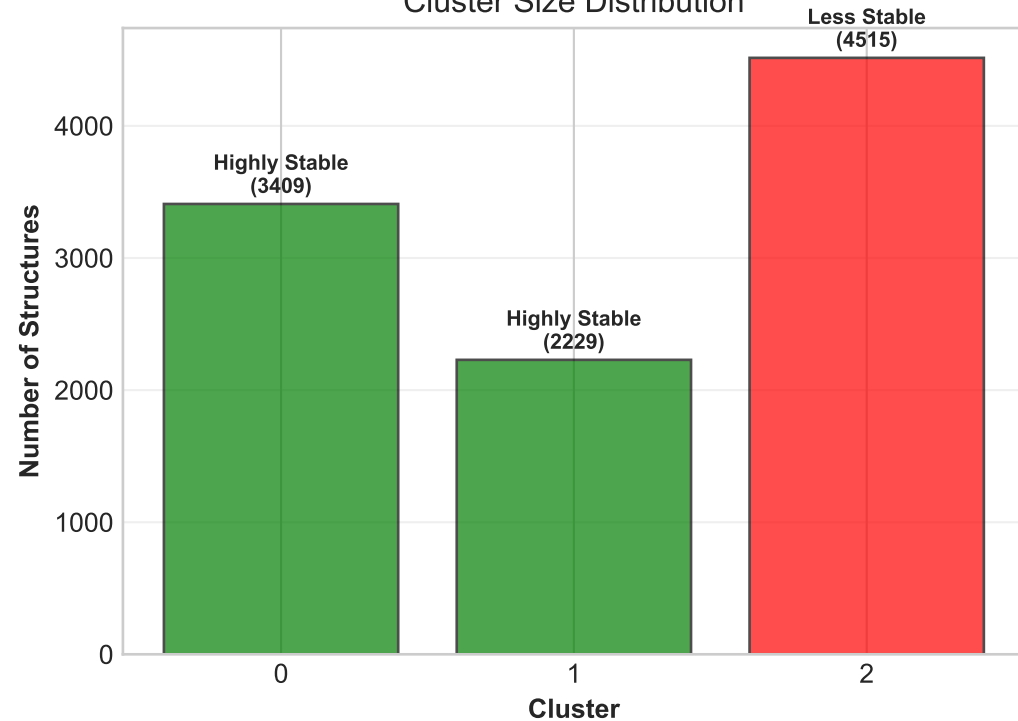
- num_atoms: 1.0000
- bond_length_range: 0.0000
- bond_complexity: 0.0000

IMPLICATIONS

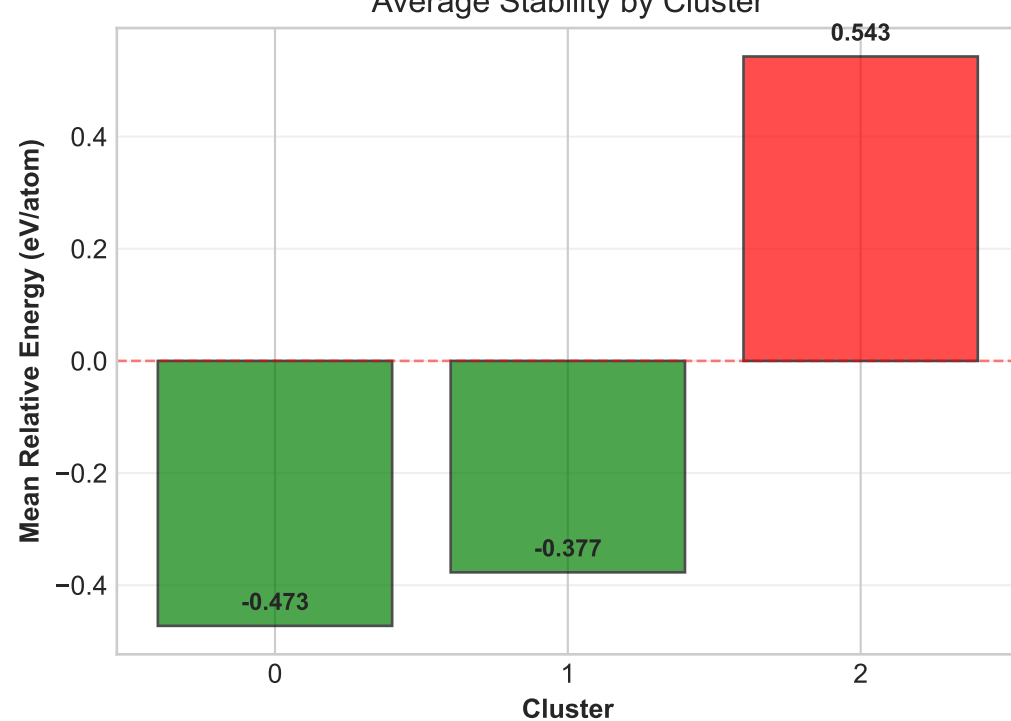
- Geometric features successfully predict energy stability
- Clear separation between stable and unstable structures
- Results can guide material discovery and design

Stability Analysis

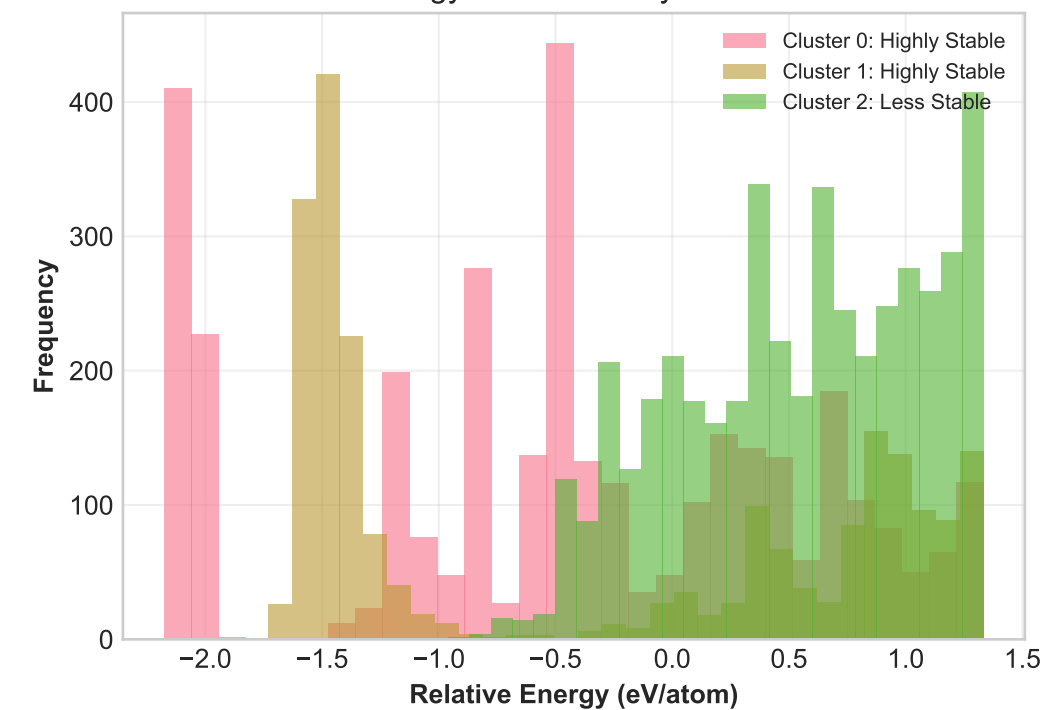
Cluster Size Distribution



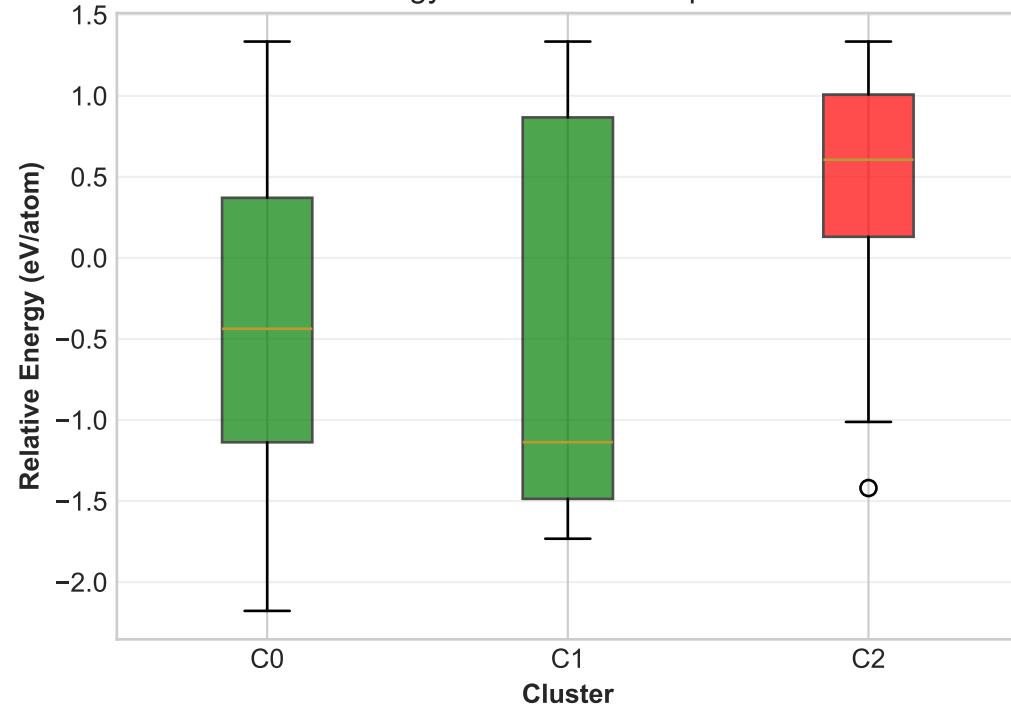
Average Stability by Cluster



Energy Distribution by Cluster

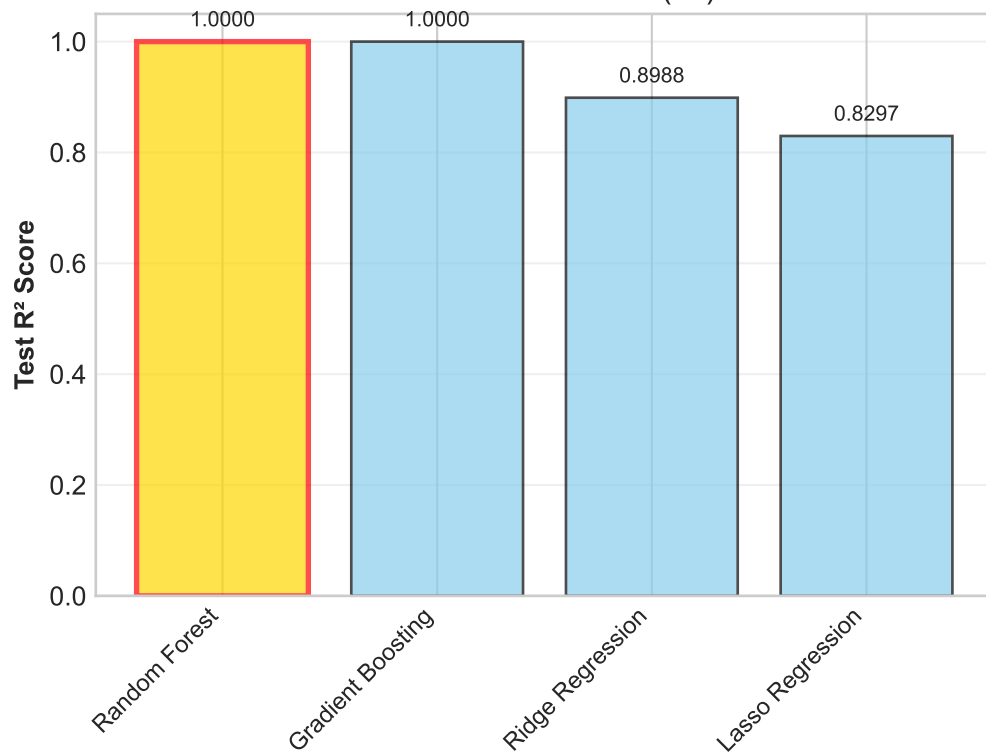


Energy Distribution Comparison

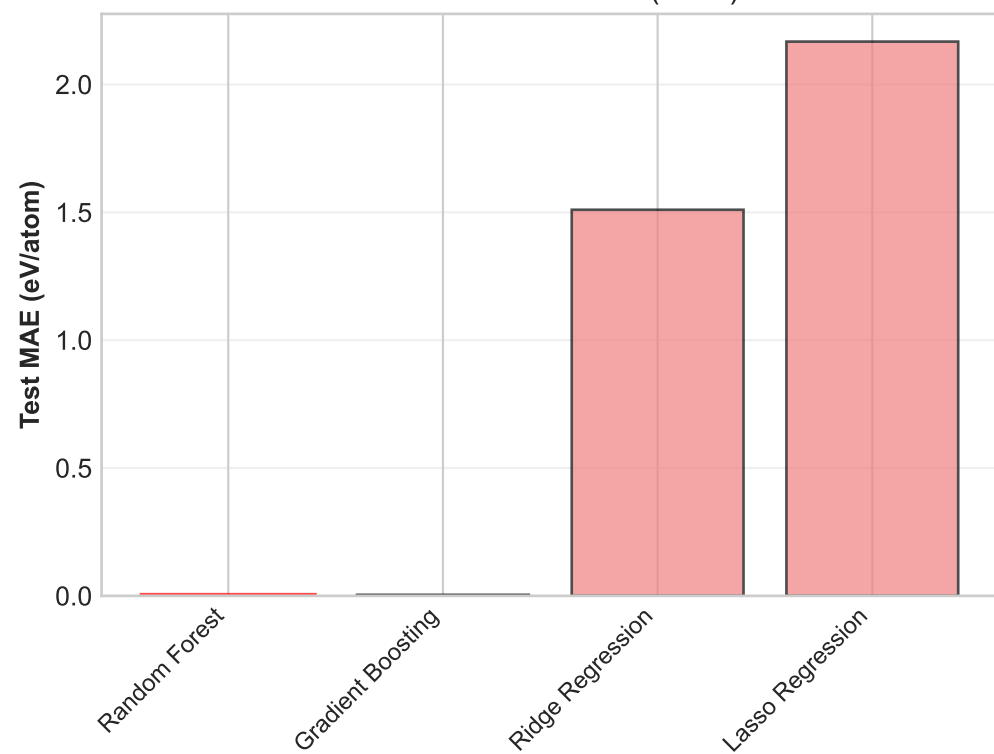


Energy Prediction Analysis

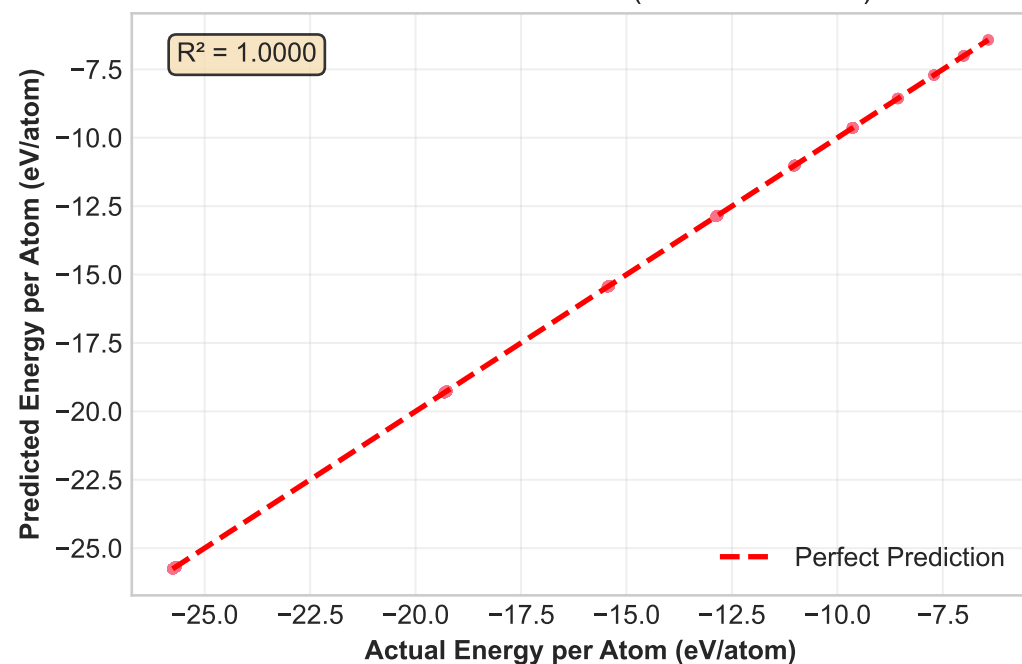
Model Performance (R²)



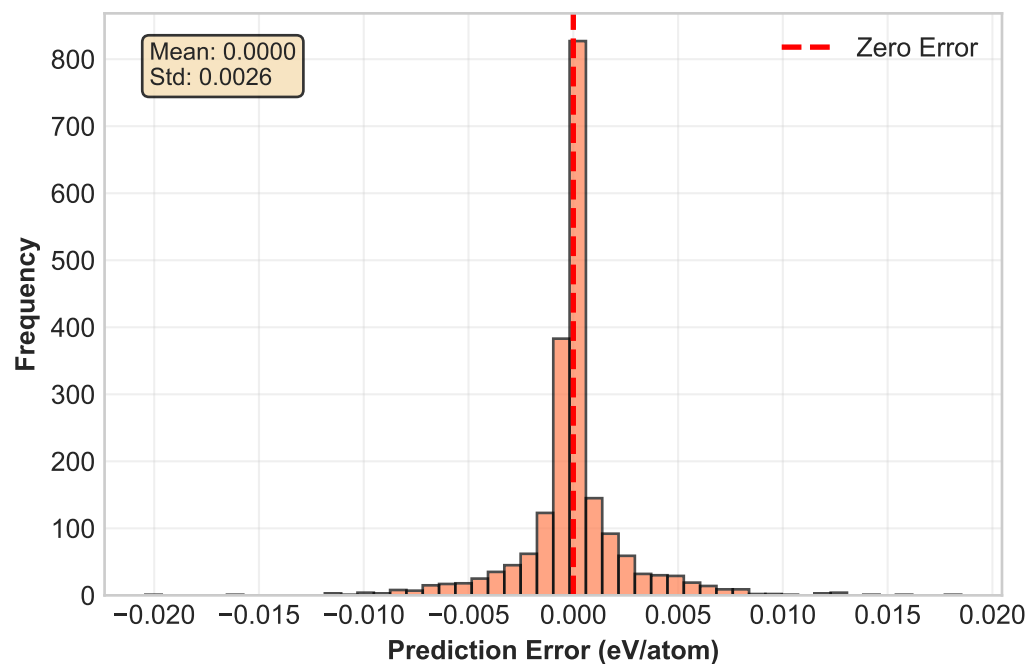
Model Performance (MAE)



Prediction Performance (Random Forest)

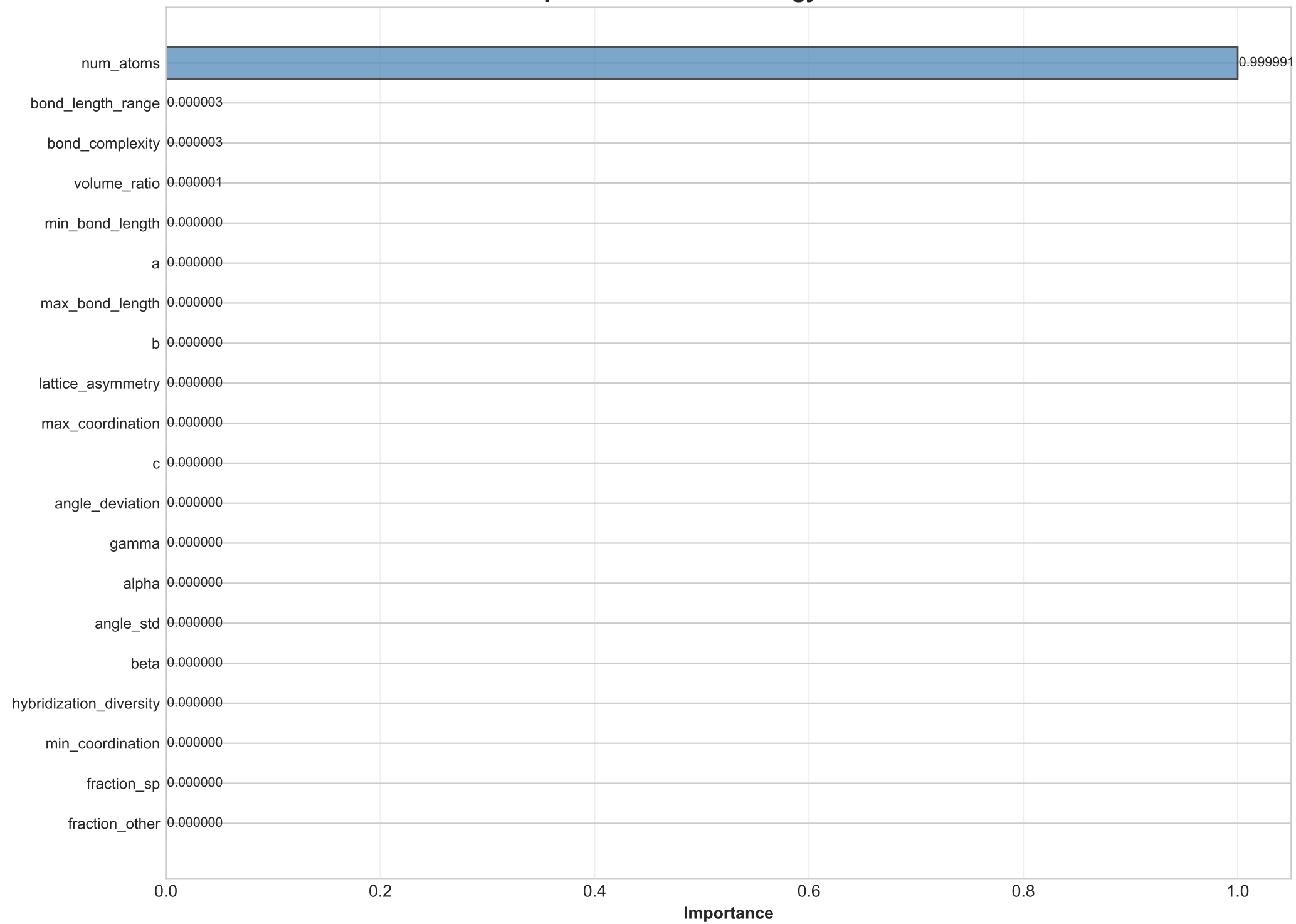


Error Distribution



Feature Importance Analysis

Top 20 Features for Energy Prediction



STABILITY STATISTICS BY CLUSTER

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Cluster 0: Highly Stable
Count: 3,409
Mean: -0.4725 eV/atom
Median: -0.4371 eV/atom
Std Dev: 1.0173 eV/atom
Min: -2.1775 eV/atom
Max: 1.3348 eV/atom
25th %ile: -1.1373 eV/atom
75th %ile: 0.3708 eV/atom

Detailed Statistics

Cluster 1: Highly Stable
Count: 2,229
Mean: -0.3771 eV/atom
Median: -1.1366 eV/atom
Std Dev: 1.1525 eV/atom
Min: -1.7323 eV/atom
Max: 1.3348 eV/atom
25th %ile: -1.4862 eV/atom
75th %ile: 0.8667 eV/atom

Cluster 2: Less Stable
Count: 4,515
Mean: 0.5429 eV/atom
Median: 0.6060 eV/atom
Std Dev: 0.5278 eV/atom
Min: -1.4196 eV/atom
Max: 1.3348 eV/atom
25th %ile: 0.1305 eV/atom
75th %ile: 1.0077 eV/atom

MODEL PERFORMANCE STATISTICS

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Random Forest
Train R²: 1.0000
Test R²: 1.0000
Train MAE: 0.0005 eV/atom
Test MAE: 0.0014 eV/atom
Train RMSE: 0.0010 eV/atom
Test RMSE: 0.0026 eV/atom
Time: 9.45 seconds

Gradient Boosting
Train R²: 1.0000
Test R²: 1.0000
Train MAE: 0.0041 eV/atom
Test MAE: 0.0042 eV/atom
Train RMSE: 0.0054 eV/atom
Test RMSE: 0.0055 eV/atom
Time: 4.10 seconds

Ridge Regression
Train R²: 0.9039
Test R²: 0.8988
Train MAE: 1.4933 eV/atom
Test MAE: 1.5101 eV/atom
Train RMSE: 1.7867 eV/atom
Test RMSE: 1.8065 eV/atom
Time: 0.01 seconds

Lasso Regression
Train R²: 0.8360
Test R²: 0.8297
Train MAE: 2.1481 eV/atom
Test MAE: 2.1678 eV/atom
Train RMSE: 2.3339 eV/atom
Test RMSE: 2.3428 eV/atom
Time: 0.00 seconds